Semidefinite Programming (SDP) & Maximum Stable Set (Lec13.pdf)

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Todays topics & Logistics

- **S**emi**D**efinite **P**rogramming (SDP)
  - Definitions & basic properties
- Examples & CVX
- LP special case
- Applications to Combinatorial Optimization
  - Maximum stable set
  - The Lovasz SDP

SDP

- **Definition**: An SDP is an optimization problem of the form:

$$\boxed{\begin{array}{ll} min & Tr(CX) \\ s.t. & Tr(A_i X) = b_i, i = 1, \dots, m \\ & X \succcurlyeq 0 \end{array}}$$

- **Data**: $C \in S^{n\times n}$, $A_i \in S^{n\times n}$, $b_i \in R$, $i = 1, \dots, m$
- $S^{n\times n}$ is the set of Symmetric $n \times n$ real matrices: $S^{n\times n} \subset R^{n\times n}$
- $Tr(H)$ is the **Trace** of the square matrix $H \in S^{n\times n}$:
  - $Tr(H) = \sum_{i=1}^{n} h_{ii}$ → **the sum of the diagonal elements**

SDP

- Why do we need the TRACE notation??

- It is a convenient way to write a **"linear function" in terms of the elements of the matrix X**

$$Tr(CX) = \sum_{i,j=1}^{n} c_{ij} x_{ij}$$

Example 1: $H = CX = \begin{pmatrix} c_{11} & c_{12} \\ c_{12} & c_{22} \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} \\ x_{12} & x_{22} \end{pmatrix} = \begin{pmatrix} c_{11}x_{11} + c_{12}x_{12} & c_{11}x_{12} + c_{12}x_{22} \\ c_{12}x_{11} + c_{22}x_{12} & c_{12}x_{12} + c_{22}x_{22} \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix}$

$$Tr(H) = Tr(CX) = (c_{11}x_{11} + c_{12}x_{12}) + (c_{12}x_{12} + c_{22}x_{22}) = h_{11} + h_{22}$$

SDP

- **Example 2:** consider the inequality: $4X_{11} - 6X_{12} + 5X_{23} \leq 7$
  - How can we write it in the form: $Tr(AX) \leq b$??
  - We need to find: b, A, X

SDP

- **Example 2:** consider the inequality: $4x_{11} - 6x_{12} + 5x_{23} \leq 7$
  - How can we write it in the form: $Tr(AX) \leq b$??
  - We need to find: b, A, X
  - $b = 7$, $A = \begin{pmatrix} 4 & -3 & 0 \\ -3 & 0 & 5/2 \\ 0 & 5/2 & 0 \end{pmatrix}$, $X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{12} & x_{22} & x_{23} \\ x_{13} & x_{23} & x_{22} \end{pmatrix}$

SDP → application in Statistics and Finance

- **Example 3:** Find the nearest correlation matrix X to a given matrix $A \in S^{n\times n}$
  - All the elements of $\boldsymbol{X}$ are **unknown** but X has some important properties

- $$\begin{aligned} min \quad & \frac{1}{2}\left|\left|A - X\right|\right|_F^2 \\ s.t. \quad & diag\{X\} = \mathbf{1} \\ & X \in S^{n\times n} \\ & X \succcurlyeq 0 \end{aligned}$$

Objective: Minimize the distance from the known matrix A

Constraints: List all the properties of a correlation matrix

Frobenius norm: $\left|\left|C\right|\right|_F = \sqrt{\sum\sum\left|c_{ij}\right|^2} = \left(Tr(C^TC)\right)^{1/2}$

Example: In portfolio optimization, we want to find a valid correlation matrix that reflects the relationships between asset returns while minimizing the deviation from a given matrix.

Review on PSD matrices

- **Definition**: Let $X \in S^{n\times n}$. $X \succcurlyeq 0 \Leftrightarrow \mathrm{y}^{\mathrm{T}}\mathrm{Xy} \geq 0, \forall y \in R^n$

- **Theorems**:

  - $X \succcurlyeq 0 \Leftrightarrow$ all eigenvalues of X are ≥ 0

  - $X \succcurlyeq 0 \Leftrightarrow$ all principal minors of X are ≥ 0 (Sylvester criterion)

  - $\boldsymbol{X \succcurlyeq 0 \Leftrightarrow X = MM^T}$ **for some matrix** $\boldsymbol{M \in R^{n\times m}}$ **(factorization of** $\boldsymbol{X}$**)**

    - Cholesky factorization: M is an upper triangular matrix

    - Eigen-decomposition: $X = Q\Lambda Q^T = Q\Lambda^{1/2}\Lambda^{1/2}Q^T = \left(Q\Lambda^{1/2}\right)\left(Q\Lambda^{1/2}\right)^T = MM^T$,

Example of SDP in 2 vars

- Is this an SDP??

$$\begin{array}{ll} \textit{min} & x + y \\ s.t. & x + y \leq 3 \\ & \begin{pmatrix} x & 1 \\ 1 & y \end{pmatrix} \succcurlyeq 0 \end{array}$$

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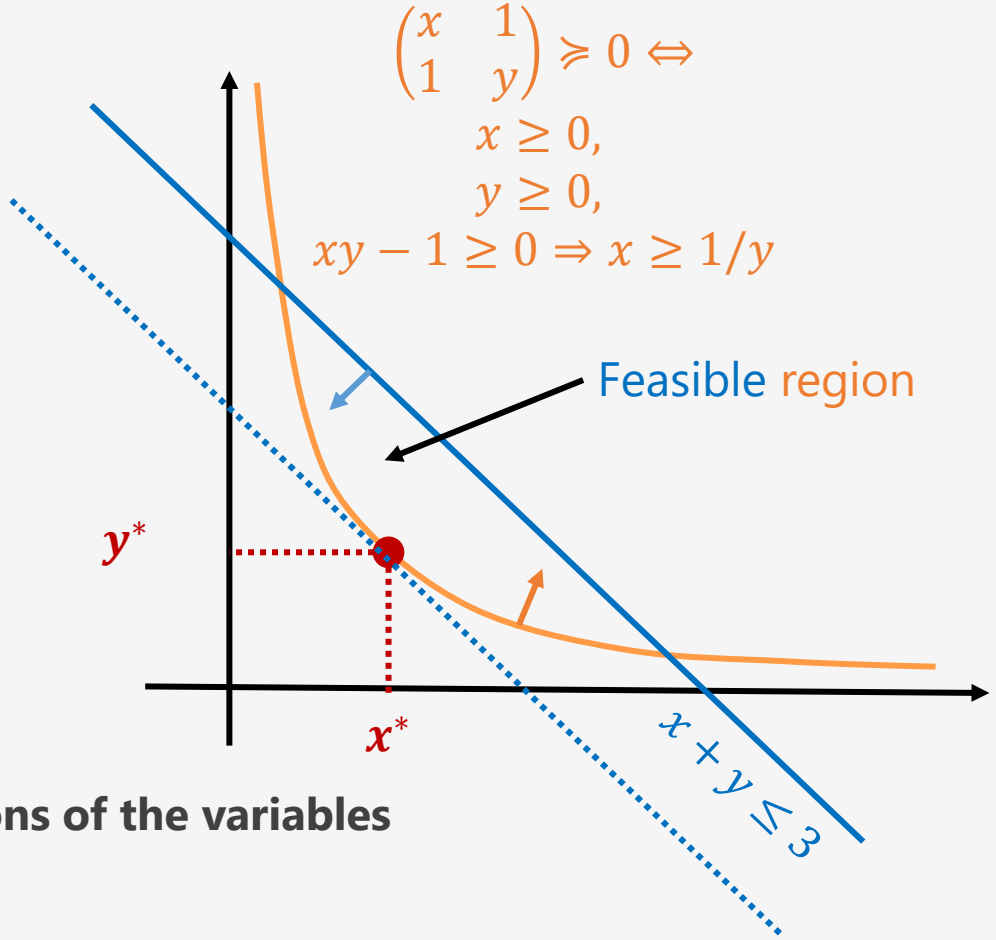


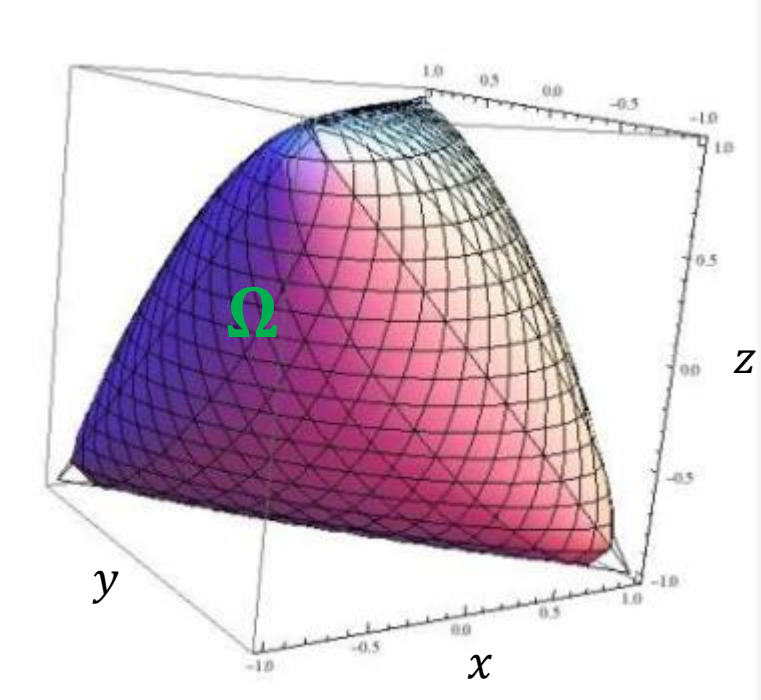
- YES because:
  - There is at least one **PSD constraint** on some matrix
  - Some of the entries of that matrix involve **affine functions of the variables**
  - **Objective function is linear**

The feasible region of an SDP is called: **Spectrahedron**

Example of a Spectrahedron in 3D

- The set Ω defines a **Spectrahedron** in 3D

$$\Omega = \left\{ (x, y, z) \in R^3 \mid \begin{pmatrix} 1 & x & y \\ x & 1 & z \\ y & z & 1 \end{pmatrix} \succcurlyeq 0 \right\}$$



From Lec13.pdf

The Spectrahedron and the Convex set

- **Theorem**: A spectrahedron is a convex set

- **Proof**: A spectrahedron is defined by the two types of constraints:

1. $Tr(AX)\ [\leq, =, \geq]\ b$ → it is and affine constraint in terms of X_{ij} →

   → defines a Halfspace → **convex set**

2. $X \succcurlyeq 0$ → Reminder: What set is the set of PSD matrices??

   - Hence $X \succcurlyeq 0$ defines a **convex set** (however, it is **not an affine constraint!**)

The Spectrahedron and the Convex set

- **Question**: Is an SDP problem a Convex Optimization problem ??
- Recall how a Convex Optimization problem is defined ??

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  - Convex function for **objective** , Convex functions for $\leq$ **constraints** , Affine functions for **=** **constraints**

The Spectrahedron and the Convex set

- **Question**: Is an SDP problem a Convex Optimization (CO) problem ??
- Recall how a Convex Optimization problem is defined ??
  - Convex function for **objective** , Convex functions for $\leq$ **constraints** , Affine functions for **=** **constraints**
- In an SDP problem we have:
  - Convex Objective
  - The $\leq$ **constraints** are affine in terms of the X_{ij}s
  - The **constraint** $\boldsymbol{X \succcurlyeq 0}$ is not affine or convex per our definition of CO problem
    - Because we would need an ∞ number of affine constraints to express it

The Spectrahedron and the Convex set

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- In an SDP problem we have:
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Hence an SDP is not (per our definition) a Convex Optimization problem

But they are very close to be classified as Convex Optimization problems

CVX solver solves large-scale SDP problems very efficiently

Example of an SDP in CVX

$$A_1 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 3 & 7 \\ 1 & 7 & 5 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 2 & 8 \\ 2 & 6 & 0 \\ 8 & 0 & 4 \end{pmatrix}, \quad \text{and} \quad C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 9 & 0 \\ 3 & 0 & 7 \end{pmatrix} \qquad X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix}$$

$$\begin{aligned} \text{minimize} \quad & x_{11} + 4x_{12} + 6x_{13} + 9x_{22} + 0x_{23} + 7x_{33} \\ \text{s.t.} \quad & \\ & x_{11} + 0x_{12} + 2x_{13} + 3x_{22} + 14x_{23} + 5x_{33} = 11 \\ & 0x_{11} + 4x_{12} + 16x_{13} + 6x_{22} + 0x_{23} + 4x_{33} = 19 \\ & X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} \succeq 0. \end{aligned}$$

$$\begin{aligned} min \quad & Tr(CX) \\ s.t. \quad & Tr(A_i X) = b_i, i = 1,2 \\ & X \succcurlyeq 0 \end{aligned}$$

Matlab

```
A1=[1 0 1; 0 3 7; 1 7 5];
A2=[0 2 8; 2 6 0; 8 0 4];
C=[1 2 3; 2 9 0; 3 0 7];
n=size(C,1);
cvx_begin
    variable X(n,n) symmetric;
    minimize trace(C*X)
    subject to
        trace(A1*X) == 11;
        trace(A2*X) == 19; % alt: sum(sum(A2.*X)) == 19
        X == semidefinite(n);
cvx_end
```

SDPs are generalizations of LPs

- **Every LP can be written as an SDP:** $\text{set}\{LP\} \subset set\{SDP\}$

LP

$$\begin{aligned} min \quad & \sum_{i=1}^{n} c_i x_i \\ s.t. \quad & a_i^T x = b_i, \ i = 1, \dots, m \\ & x_j \geq 0, \ j = 1, \dots, n \end{aligned}$$

SDP

$$\begin{aligned} min \quad & Tr(CX) \\ s.t. \quad & Tr(A_i X) = b_i, i = 1, \dots, m \\ & X \succcurlyeq 0 \end{aligned}$$

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- **Question:** How can we write the LP as an SDP ???

- $C = diag(c_1, \dots, c_n)$, $A_i = diag(a_{i1}, \dots, a_{in})$, $X = diag(x_1, \dots, x_n)$

- $c^T x = Tr(CX)$, $a_i^T x = b_i \Leftrightarrow Tr(A_i X) = b_i$, $x_j \geq 0, \ j = 1, \dots, n \Leftrightarrow X \succcurlyeq 0$

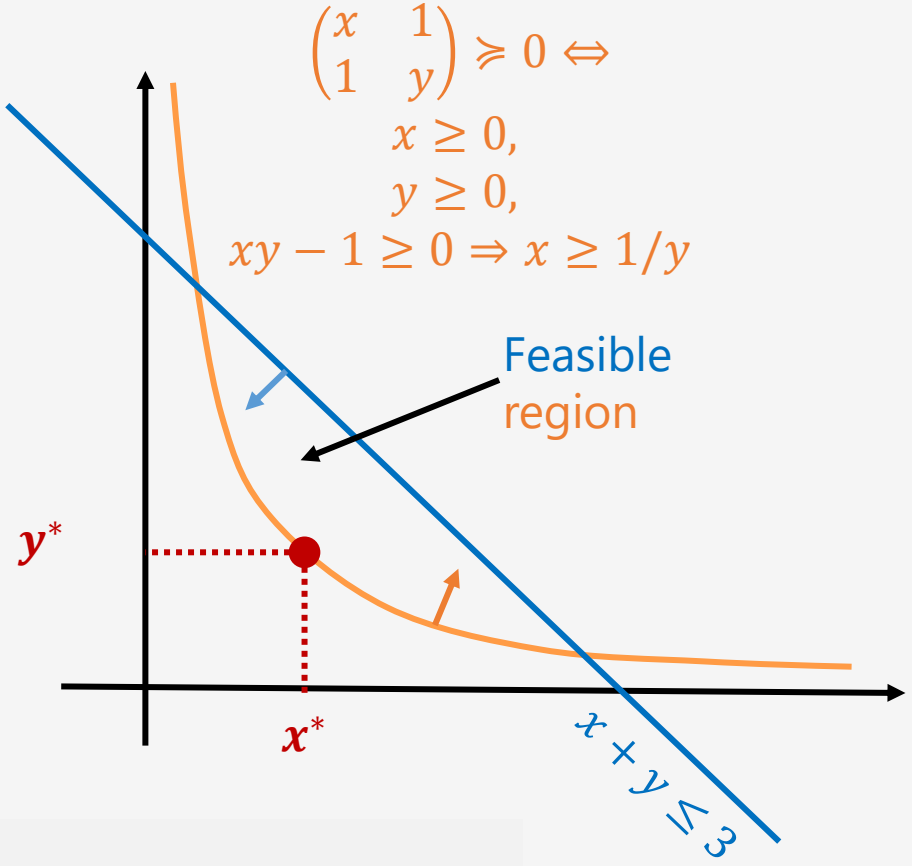
SDPs are generalizations of LPs

- But not every SDP can be written as an LP!

- Recall this SDP:

$$\begin{array}{ll} min & x+y \\ s.t. & x+y \leq 3 \\ & \begin{pmatrix} x & 1 \\ 1 & y \end{pmatrix} \succcurlyeq 0 \end{array}$$

- **Question:** Why can't we write it as an LP??



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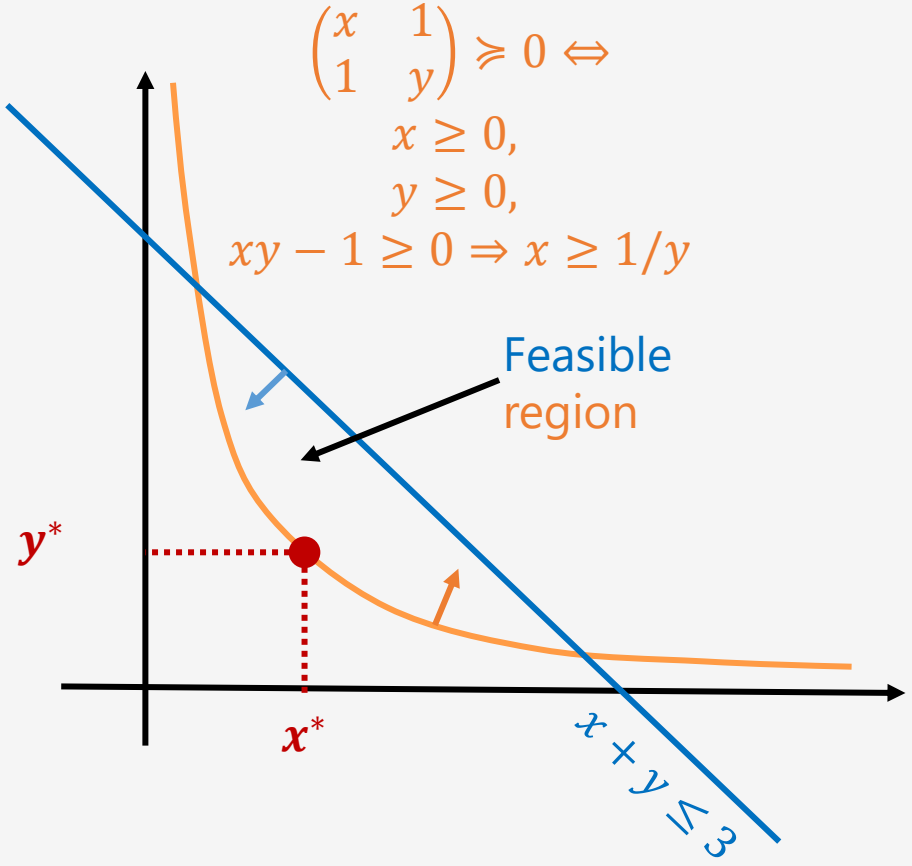
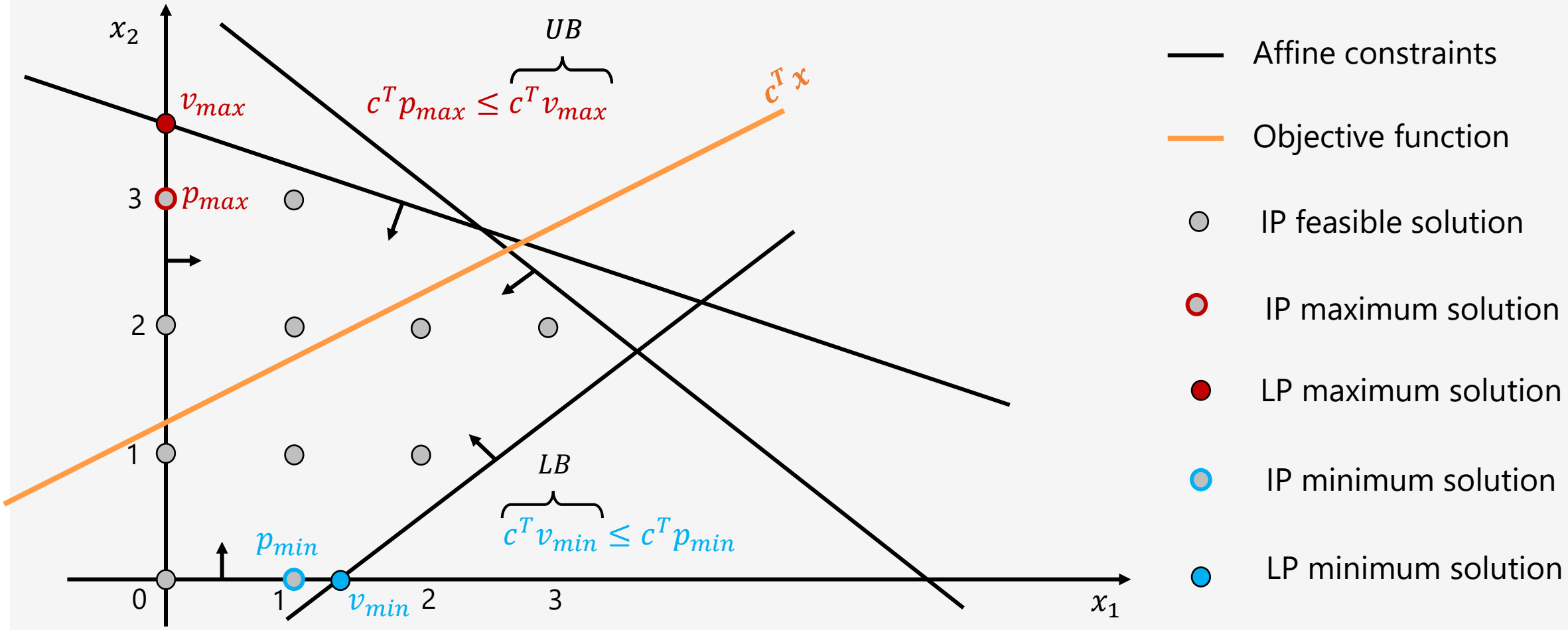


- **Question:** Why can't we write it as an LP??
  - We would need an ∞ **number of affine constraints** to express the ∞ **number of extreme point** of the PSD constraint

Lower and Upper Bounds based on LP relaxations of IP problems



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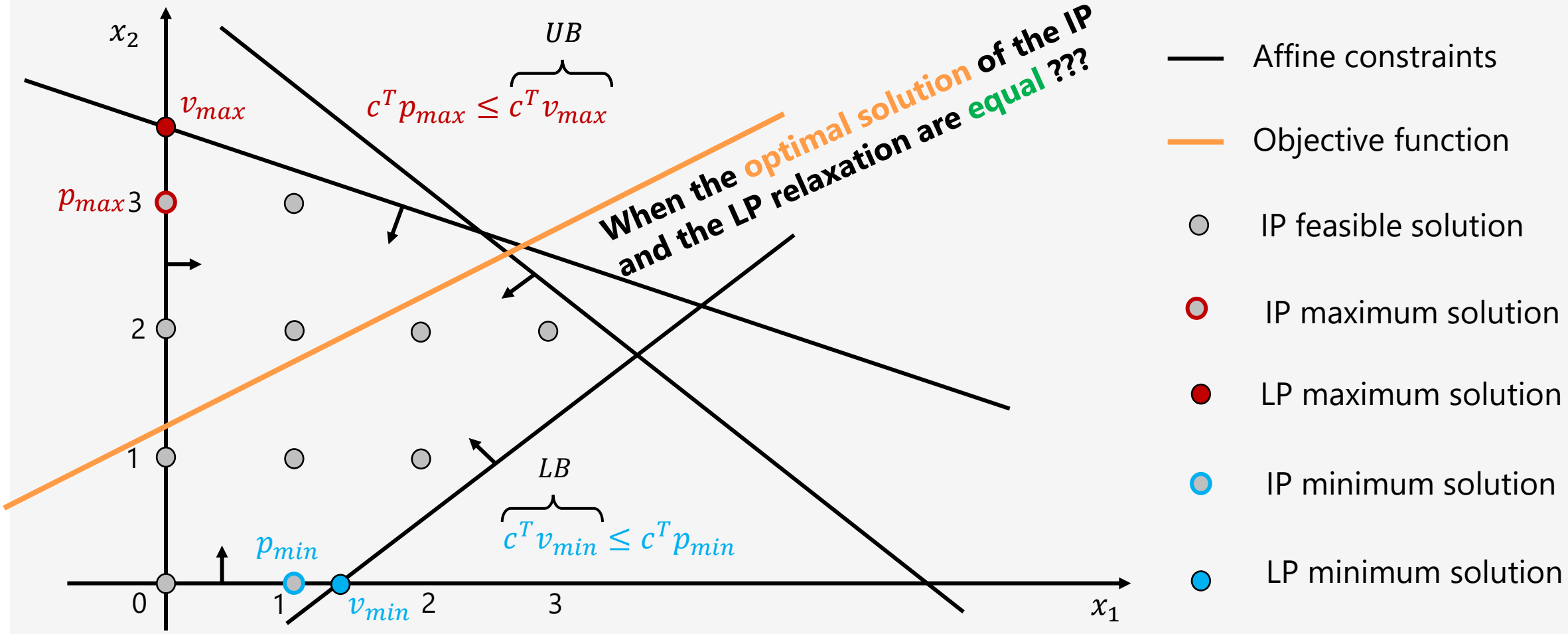

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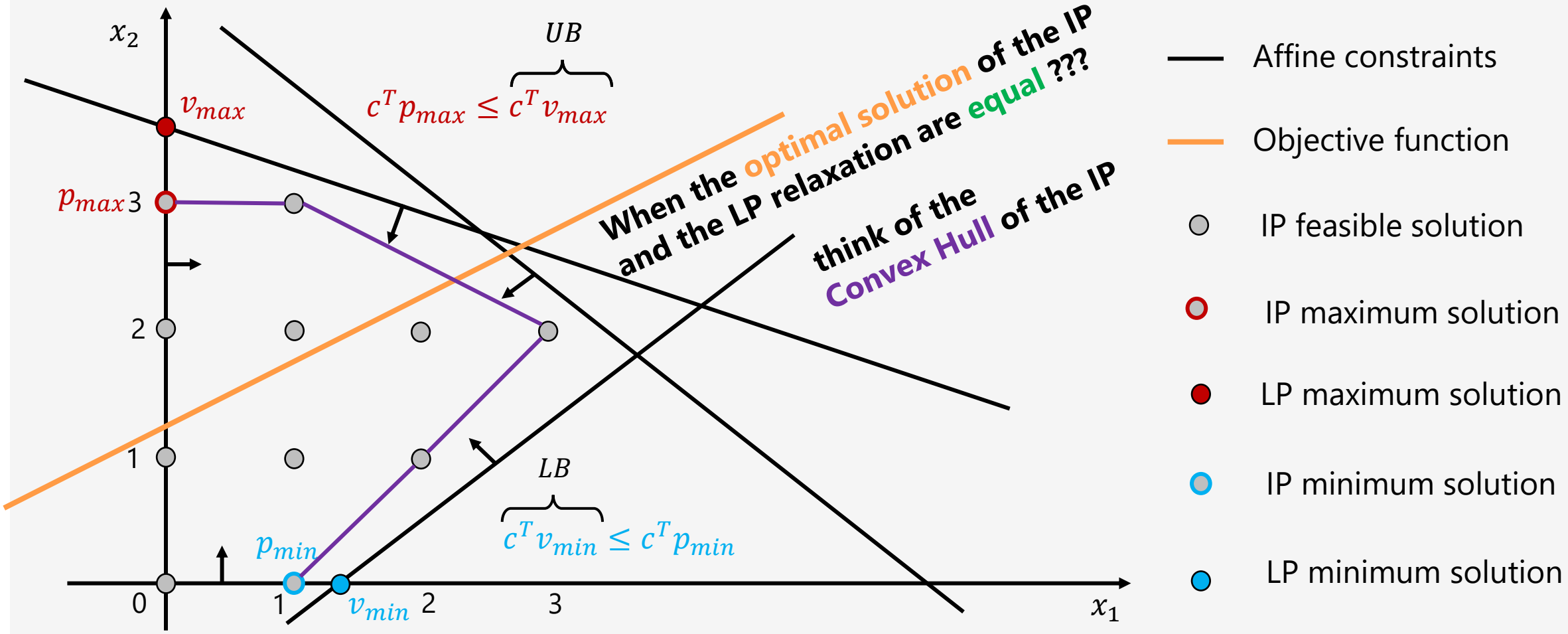
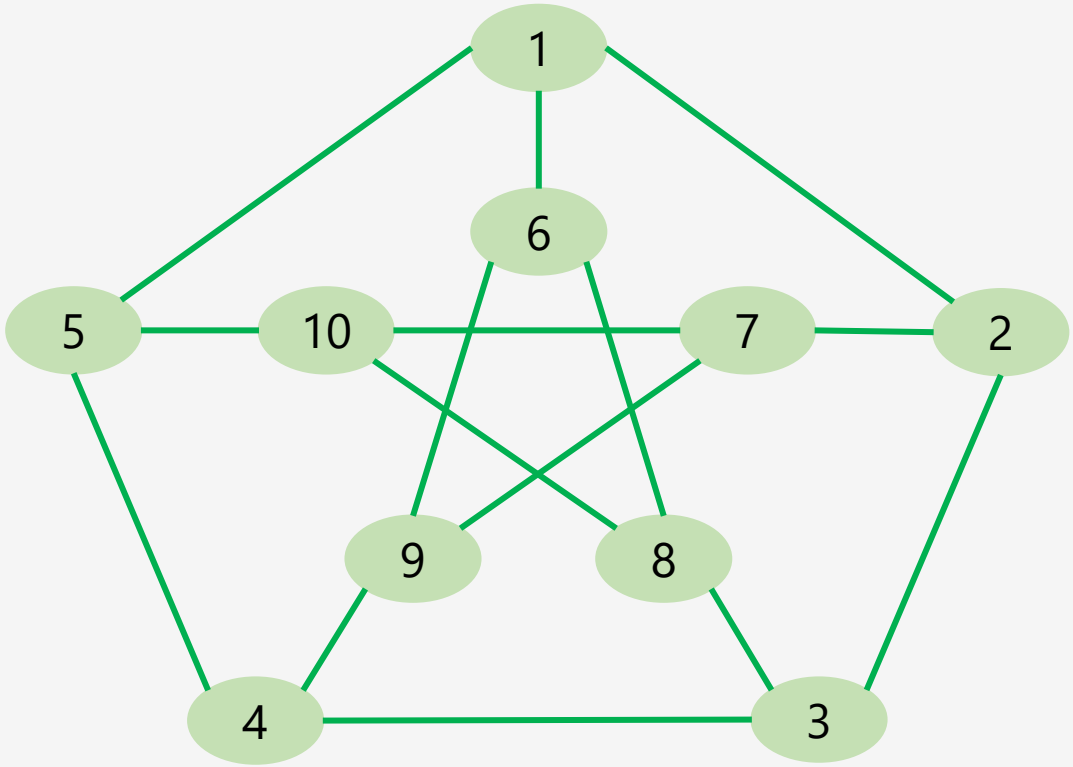


Applications of SDP in Combinatorial Optimization

- Find the **largest stable (independent) set** of a Petersen graph: $G = (V, E)$



$$V = \{1,2, \dots, 10\}$$
$$G = \{(1,2), (2,3), \dots, (7,10)\}$$

Recall: A stable set in a graph is a set of pairwise disjoint (not connected) nodes

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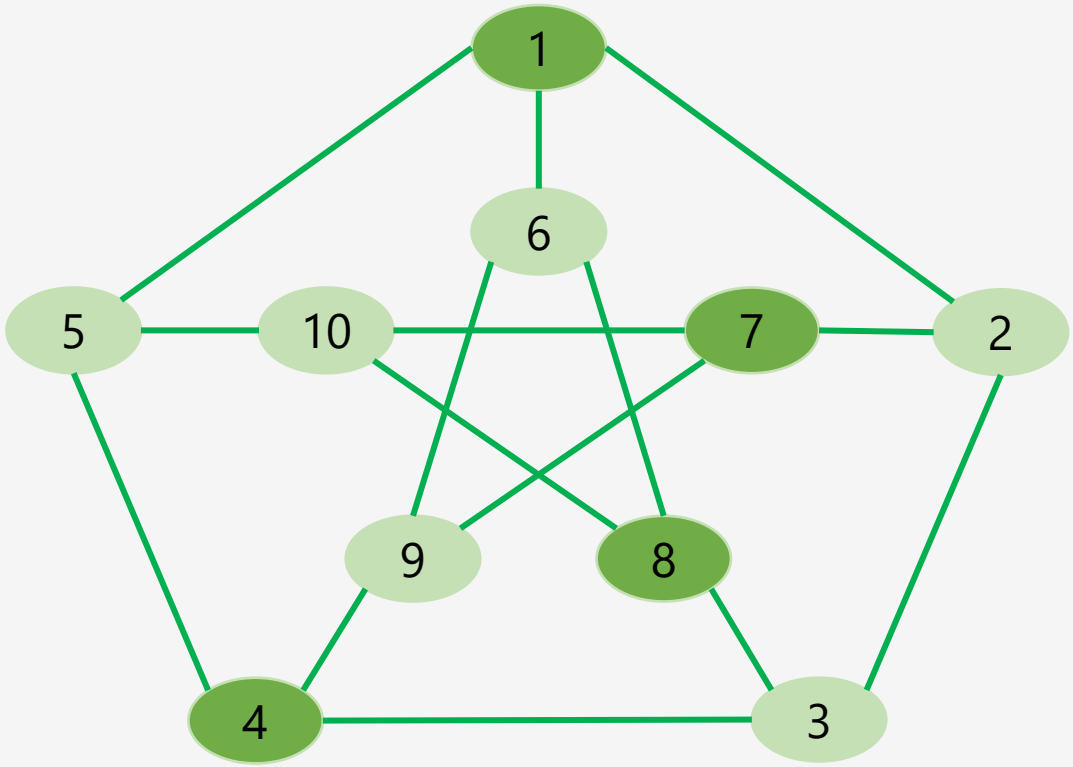

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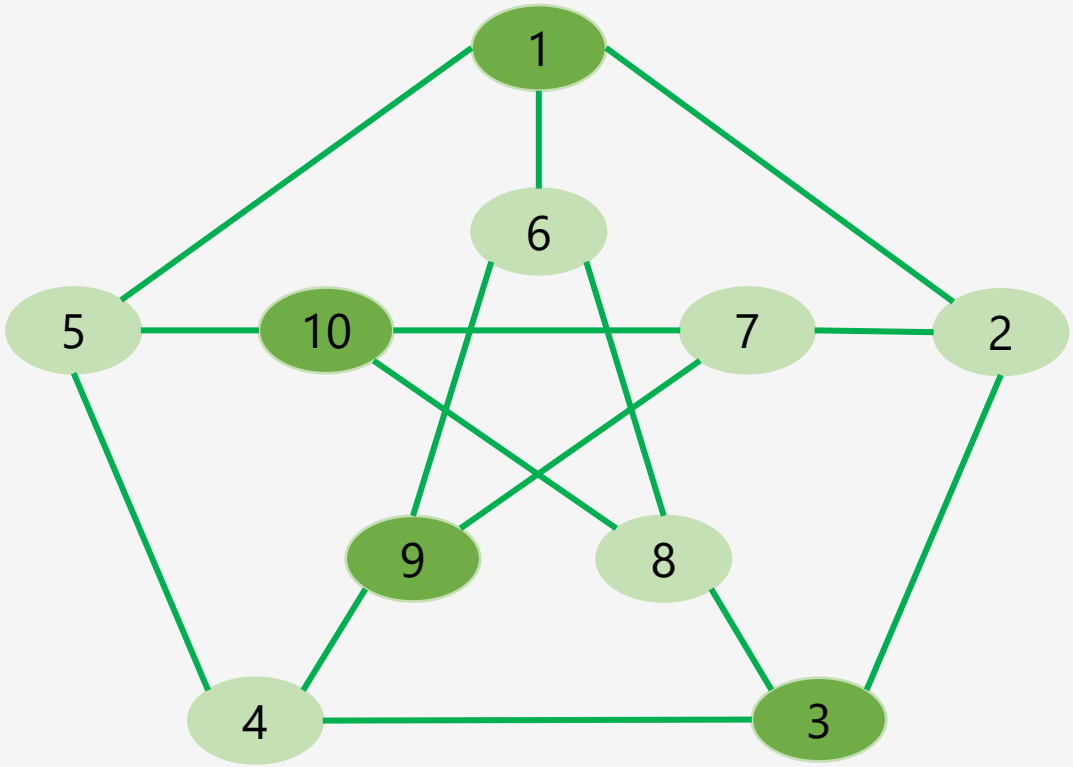

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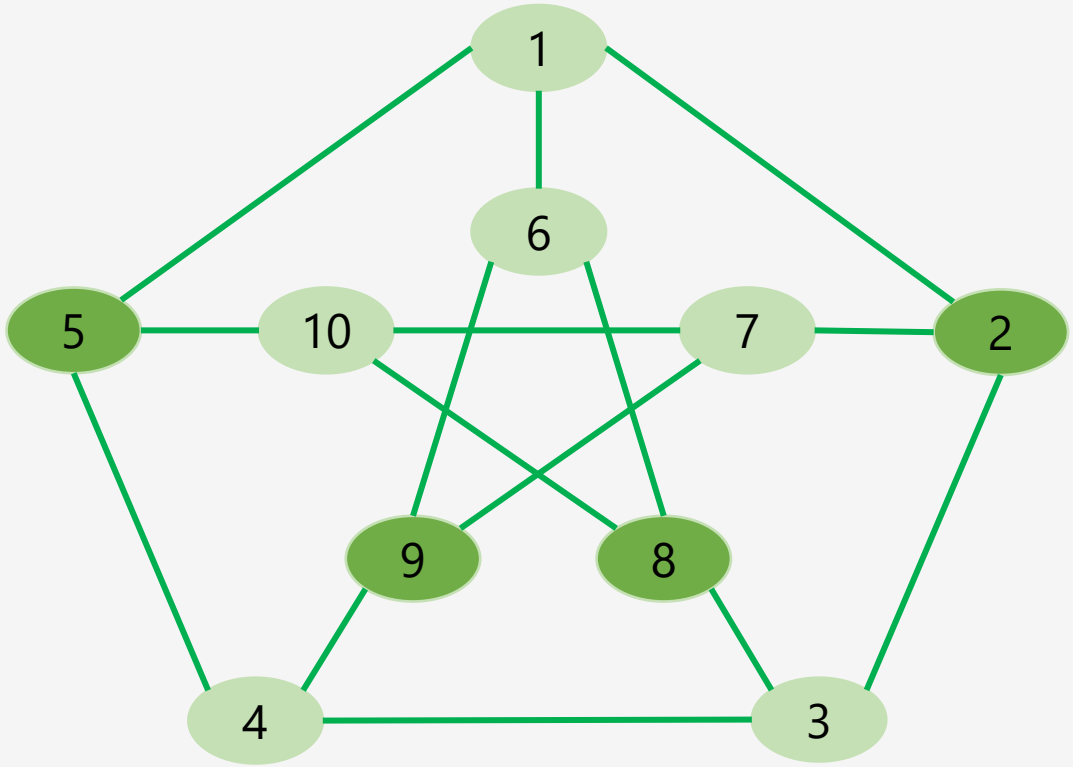

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Applications of SDP in Combinatorial Optimization

- From the previous slides **we believe** that the largest stable set has size 4
- Hence, we can say that 4 is a **lower bound** of $\alpha(G)$, that is: $4 \leq \alpha(G)$
- **Question:** Can we find an **upper bound** ??
  - **Hint**: what type of optimization problem we used to formulate the largest stable set ??

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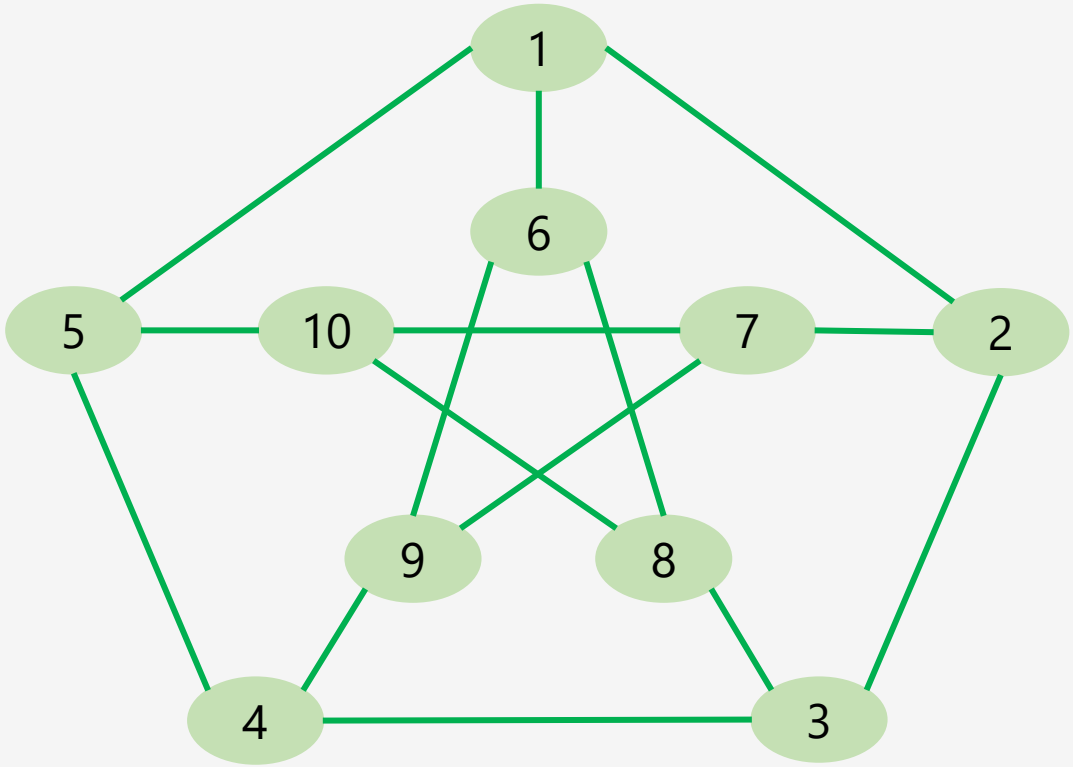
- An **Integer Programming problem** (aka, Combinatorial Optimization problem)

Applications of SDP in Combinatorial Optimization

Find the largest stable set of a Petersen graph: $G = (V, E)$

$$V = \{1,2,\dots,10\}$$
$$G = \{(1,2),(2,3),\dots,(7,10)\}$$



- **Decision variables:**
  - $x_i = 1$, if node i is picked;
  - $x_i = 0$, otherwise
- **Objective:**
  - $\mathbf{max}\, f(x) = \sum_{i=1}^{10} x_i$
- **Constraints:**
  - $x_i + x_j \leq 1, \forall (i,j) \in E$
  - $\boldsymbol{x_i} \in \{\mathbf{0}, \mathbf{1}\}, \forall i \in V$ → binary constraint

How can we find an **upper bound** of this IP ??

Applications of SDP in Combinatorial Optimization

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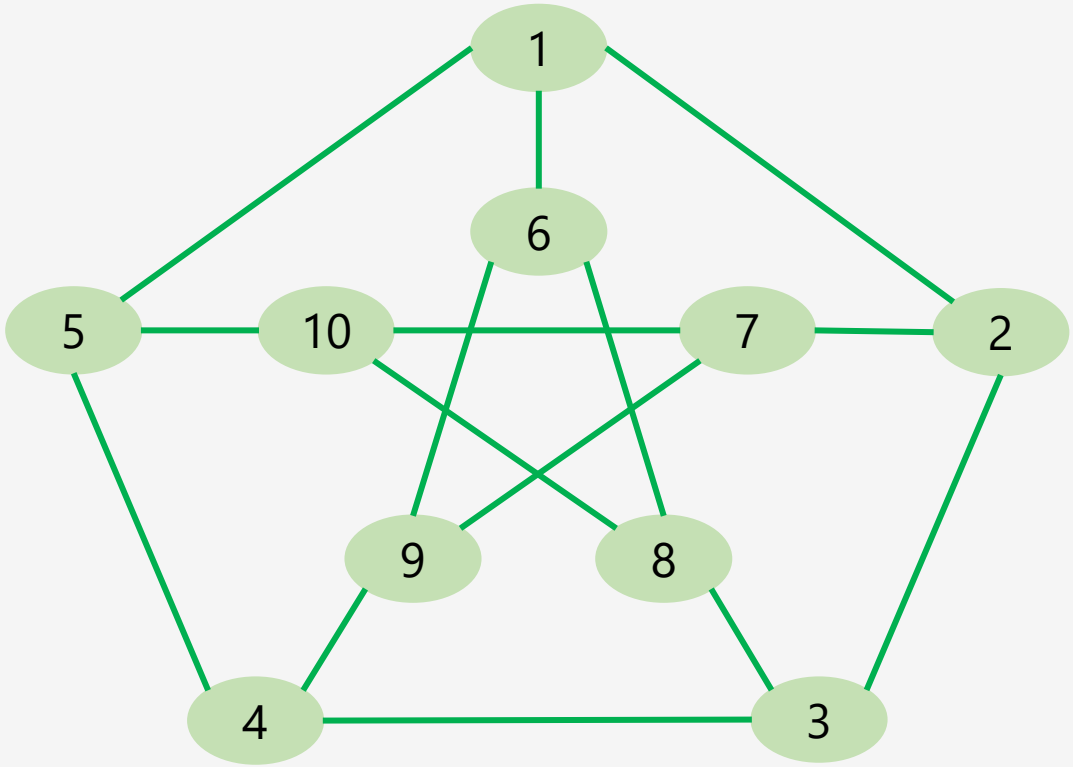

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  - $x_i + x_j \leq 1, \forall (i,j) \in E$
  - $\mathbf{0 \leq x_i \leq 1}$, $\forall i \in V$ → continuous constraints

How can we find an **upper bound** of this IP ??

Considering its **LP (continuous) relaxation**!

Applications of SDP in Combinatorial Optimization

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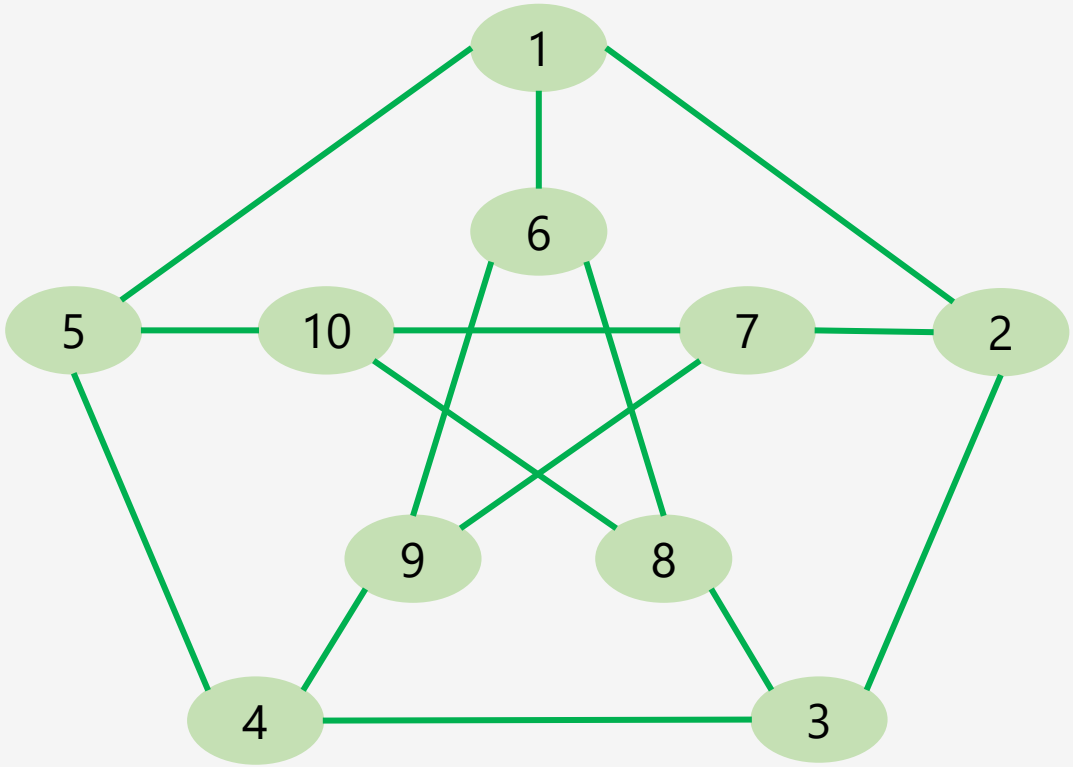

- $LP_{OPT} = \mathbf{max}\ \ f(x) = \sum_{i=1}^{10} x_i$
  s.t. $x_i + x_j \leq 1, \forall (i,j) \in E$
  $0 \leq x_i \leq 1, \forall i \in V$

$$\alpha(G) \leq LP_{OPT}$$

But LP_{OPT} can be **much larger than** $\alpha(G)$

This is <u>not a good-quality Upper Bound!</u>

Can we find an UB that is as close as possible to $\alpha(G)$??

Applications of SDP in Combinatorial Optimization

- We can create an SDP whose optimal solution is much closer to $\alpha(G)$ than the LP relaxation:

  - $$SDP_{OPT} = \max \quad Tr(JX)$$
    $$\text{s.t.} \quad Tr(X) = 1$$
    $$X_{ij} = 0, \quad \forall (i,j) \in E$$
    $$X \succcurlyeq 0$$

Lovasz SDP

  - where $J = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}$

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For the Petersen graph we have: $SDP_{OPT} = 4$

- where $J = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}$

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For the Petersen graph we have: $SDP_{OPT} = 4$

Recall that for the Petersen graphs: $\alpha(G) = 4$

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For the Petersen graph we have: $SDP_{OPT} = 4$

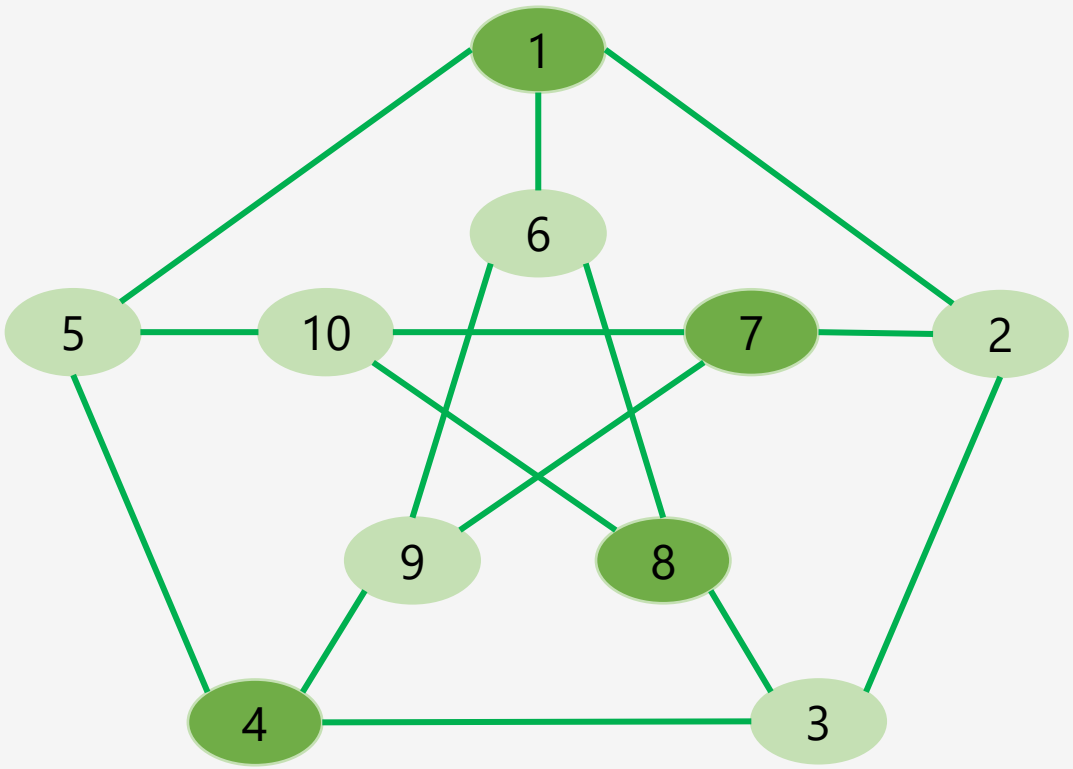
Recall that for the Petersen graphs: $\alpha(G) = 4$

Hence: $\boldsymbol{SDP_{OPT}}$ is the best UB we can get because

$$\boldsymbol{\alpha(G) = SDP_{OPT}}$$

Applications of SDP in Combinatorial Optimization

- **Theorem**: The optimal solution SDP_{OPT} is an **upper bound** of $\alpha(G)$, that is $\alpha(G) \leq SDP_{OPT}$
- **Proof**: Let $\hat{x} \in \{0,1\}^n$ be the indicator vector of the largest stable set $\hat{S}$ in the graph $G = (V, E)$.
- For example, for the Petersen graph we have:
  - $\hat{S} = \{1,4,7,8\}$
  - $\hat{x} = \{1,0,0,1,0\ ,0,1,1,0,0\} \in \{0,1\}^{10}$
  - Note that $\alpha(G) =$ "# nodes in $\hat{S}$"



Applications of SDP in Combinatorial Optimization

- **Proof (cont):** Consider the matrix $\boldsymbol{\hat{X}} = \frac{\mathbf{1}}{\boldsymbol{\alpha(G)}}\boldsymbol{\hat{x}\hat{x}^T}$

- Claim 1: $\hat{X}$ is **feasible** for the Lovasz SDP problem. <u>Need to show</u>: $\hat{X} \succcurlyeq 0$, $\hat{X}_{ij} = 0, \forall (i,j) \in E$, and $Tr(\hat{X}) = 1$

Applications of SDP in Combinatorial Optimization

- **Proof (cont):** Consider the matrix $\boldsymbol{\hat{X}} = \frac{\mathbf{1}}{\boldsymbol{\alpha(G)}}\boldsymbol{\hat{x}\hat{x}^T}$

- Claim 1:Let's see how the matrix $\boldsymbol{\hat{X}} = \frac{\mathbf{1}}{\boldsymbol{\alpha(G)}}\boldsymbol{\hat{x}\hat{x}^T}$ looks like for the Petersen graph:

$$\hat{X} = \frac{1}{\alpha(G)}\hat{x}\hat{x}^T = \frac{1}{\alpha(G)}\overset{\hat{x}}{\begin{pmatrix}1\\0\\0\\1\\0\\0\\1\\1\\0\\0\end{pmatrix}}\overset{\hat{x}^T}{\begin{pmatrix}1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0\end{pmatrix}} = \frac{1}{\alpha(G)}\begin{bmatrix}1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0\\1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

Applications of SDP in Combinatorial Optimization

- **Proof (cont):** Consider the matrix $\boldsymbol{\hat{X} = \frac{1}{\alpha(G)}\hat{x}\hat{x}^T}$

- Claim 1: $\hat{X}$ is **feasible** for the Lovasz SDP problem. <u>Need to show</u>: $\hat{X} \succcurlyeq 0$, $\hat{X}_{ij} = 0, \forall (i,j) \in E$, and $Tr(\hat{X}) = 1$

  - $\boldsymbol{\hat{X} \succcurlyeq 0}$

    - $\forall y \in R^n$, $y^T\hat{X}y = \frac{1}{\alpha(G)}y^T\hat{x}\hat{x}^Ty = \frac{1}{\alpha(G)}(\hat{x}^Ty)^T(\hat{x}^Ty) = \frac{1}{\alpha(G)}\left|\left|\hat{x}^Ty\right|\right|^2 \geq 0$

Applications of SDP in Combinatorial Optimization

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  - $\boldsymbol{\hat{X}_{ij} = 0, \forall (i,j) \in E}$

    - Recall that $\hat{x}$ is an **indicator** vector for a stable set

    - Then we have $\boldsymbol{\hat{x}_i \hat{x}_j = 0}$ **for all** $\boldsymbol{(i,j) \in E}$

    - And hence $\boldsymbol{\hat{X}_{ij} = \frac{1}{\alpha(G)} \hat{x}_i \hat{x}_j = 0}$ for all $(i,j) \in E$

Applications of SDP in Combinatorial Optimization

- **Proof (cont):** Consider the matrix $\boldsymbol{\hat{X} = \frac{1}{\alpha(G)}\hat{x}\hat{x}^T}$

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  - $\boldsymbol{\hat{X}_{ij} = 0, \forall (i,j) \in E}$

    - Recall that $\hat{x}$ is an indicator vector for a stable set

    - Then we have $\hat{x}_i\hat{x}_j = 0$ for all $(i,j) \notin E$

    - And hence $\hat{X}_{ij} = \frac{1}{\alpha(G)}\hat{x}_i\hat{x}_j = 0$ for all $(i,j) \notin E$

  - $\boldsymbol{Tr(\hat{X}) = 1}$

    - $Tr(\hat{X}) = \sum_i \hat{X}_{ii} = \frac{1}{\alpha(G)}\sum_i \hat{x}_i\hat{x}_i = \frac{1}{\alpha(G)}\sum_i \hat{x}_i^2$
      $\frac{1}{\alpha(G)}\sum_i \hat{x}_i = \frac{1}{\alpha(G)}\alpha(G) = 1$

Applications of SDP in Combinatorial Optimization

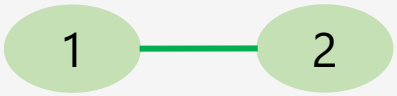
- **Proof (Cont.):**
  - **Claim 2:** We also need to show that: $\alpha(G) \leq Tr(J\hat{X})$ (**for Upper Bound**)
  - Using the following facts, you can easily show this claim (take as exercise!)
    - **Fact 1**: $\mathrm{J} = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} = ee^T$ where $e = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$
    - **Fact 2**: $\hat{X} = \frac{1}{\alpha(G)} \hat{x}\hat{x}^T$
    - **Fact 3**: $\hat{x}$ is an indicator vector and hence $\hat{x}_i \hat{x}_j = 0$ for all $(i,j) \in E$
  - Indeed: $Tr(J\hat{X}) = Tr\left(ee^T \frac{1}{\alpha(G)} \hat{x}\hat{x}^T\right) = \frac{1}{\alpha(G)} Tr(ee^T \ \hat{x}\hat{x}^T) = \frac{1}{\alpha(G)} Tr(e^T \hat{x}\hat{x}^T e) =$

$$= \frac{1}{\alpha(G)} (e^T \hat{x})^2 = \frac{1}{\alpha(G)} \left(\alpha(G)\right)^2 = \alpha(G) \qquad \blacksquare$$

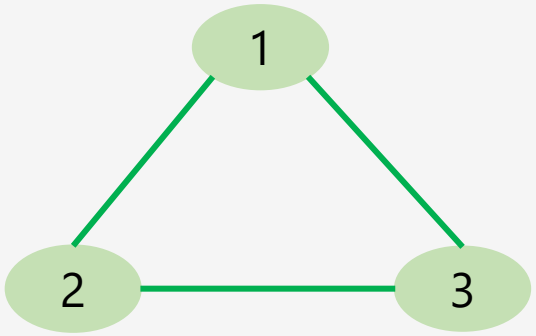
Comparison of LP_{OPT} and SDP_{OPT} in Complete Graphs

- A graph $G = (V, E)$ is called **complete** iff all nodes are connected to each other.
  - $E = \{(i, j) \mid \forall i, j \in V, i \neq j\}$
- Examples of complete graphs:

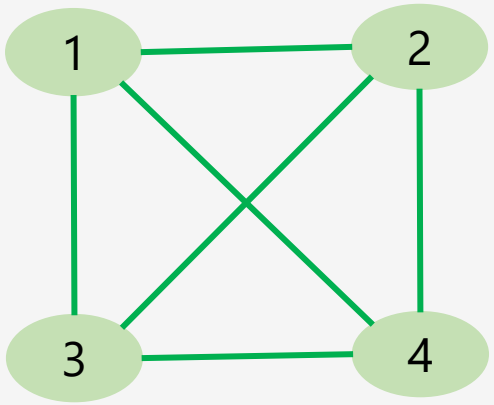
K_2



K_3



K_4



- **Question:** What is the $\alpha(G)$ for a complete graph ??

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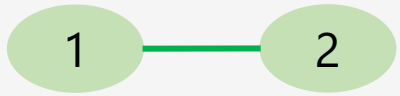
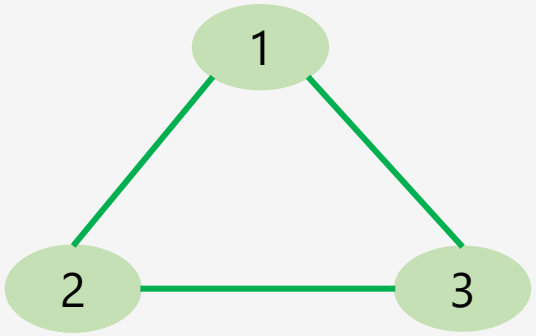
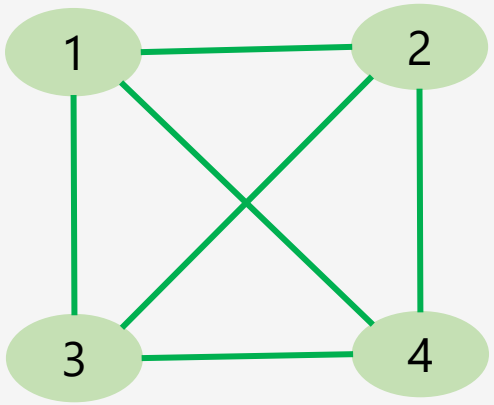


K_3



K_4



- **Question:** What is the $\alpha(G)$ for a complete graph ?? → $\boldsymbol{\alpha(G) = 1}$

Comparison of LP_{OPT} and SDP_{OPT} in Complete Graphs

- **Claim:** $\boldsymbol{LP_{OPT} \geq \frac{n}{2}}$, for every complete graph K_n
  - Recall how LP_{OPT} is defined:
  - $LP_{OPT} = \max \ f(x) = \sum_{i=1}^{n} x_i$

    s.t. $x_i + x_j \leq 1, \forall (i,j) \in E$

    $0 \leq x_i \leq 1, \forall i \in V$
  - **Question:** Can we find a feasible solution of the LP relaxation??

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  - The point $\boldsymbol{\tilde{x}} = (\frac{1}{2}.\frac{1}{2}, \dots, \frac{1}{2})$ is feasible, with objective value: $\boldsymbol{f(\tilde{x}) = \frac{n}{2}}$. Hence: $\boldsymbol{LP_{OPT} \geq \frac{n}{2}}$

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- **Claim:** $\boldsymbol{SDP_{OPT} = 1}$, for every complete graph K_n
  - Recall how SDP_{OPT} is defined:
  - $$\boldsymbol{SDP_{OPT}} = \begin{array}{ll} max & Tr(JX) \\ s.t. & Tr(X) = 1, \\ & X_{ij} = 0, \quad \forall (i,j) \in E \\ & X \succcurlyeq 0 \end{array}$$

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  - Note that for a complete graph: $X = diag(X_{11}, X_{22}, \dots, X_{nn})$, because $X_{ij} = 0, \quad \forall (i,j) \in E = V \times V$
  - Let's find the objective function value: $Tr(JX) = Tr(X) = 1$, because $JX = X$
- Recall that $\boldsymbol{\alpha(K_n) = 1}$ for all $n \geq 2$, and hence $\boldsymbol{\alpha(K_n) = SDP_{OPT}}$ **(best UB ever!)**

Duality in SDP

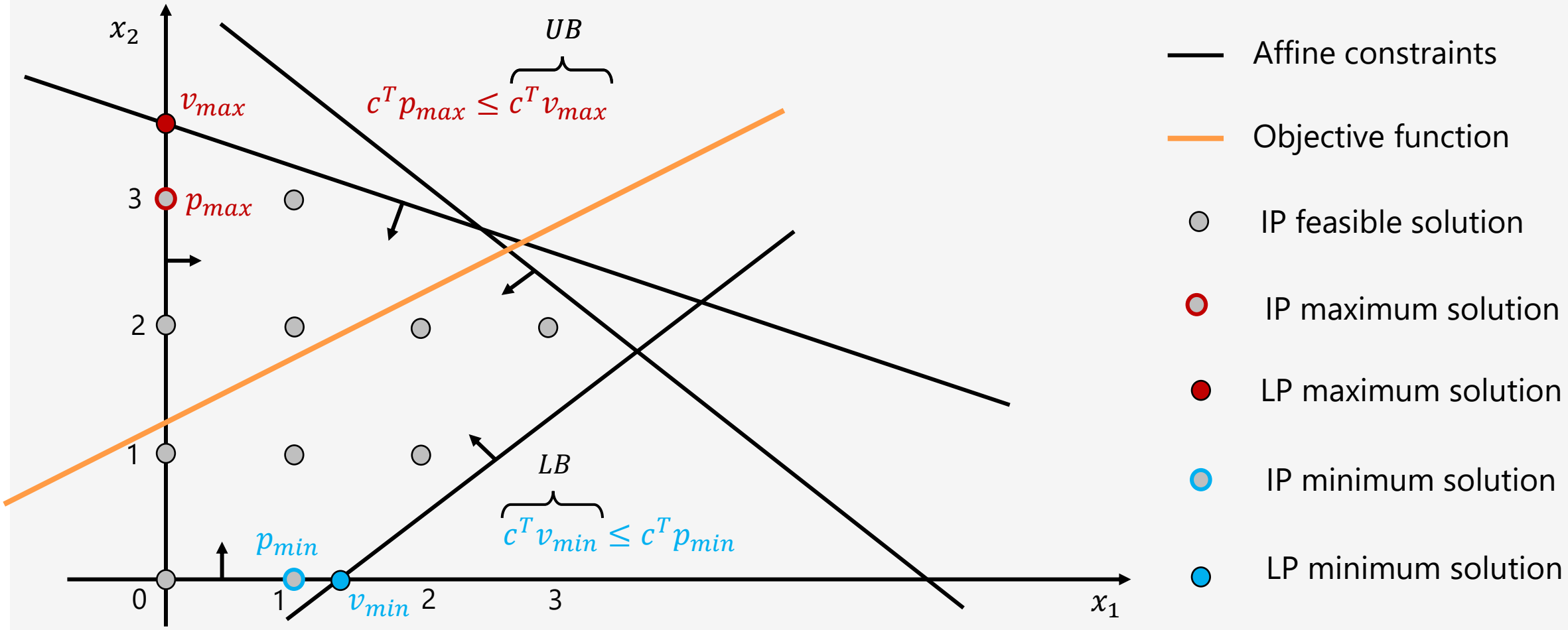
- We have similar duality definitions in SDP
- **Every SDP problem has its Dual problem**
- **Weak Duality:** if $X \in S^{n\times n}$ is feasible for the Primal problem and $y \in R^m$ is feasible for the Dual problem, then $b^T y \leq Tr(CX)$
- **Strong Duality:** if the Primal and Dual problems are both strictly feasible ($X \succ 0$ and $y > 0$), then both problems achieve their optimal values and $Tr(CX^*) = b^T y^*$

Primal problem

$$\begin{aligned} min \quad & Tr(CX) \\ s.t. \quad & Tr(A_i X) = b_i, i = 1 \dots m \\ & X \succcurlyeq 0 \end{aligned}$$

Dual problem

$$\begin{aligned} max \quad & b^T y \\ s.t. \quad & \textstyle\sum_{i=1..m} y_i A_i \preccurlyeq C \\ & y \in R^m \end{aligned}$$

.Note: $\sum_{i=1..m} y_i A_i \preccurlyeq C \iff C - \sum_{i=1..m} y_i A_i \succcurlyeq 0$

Lower and Upper Bounds based on LP relaxations of IP problems